# MATT CHAIKOWSKY

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## OBJECTIVE

Electrical Engineer pursuing an M.S. in Electrical Engineering at Rochester Institute of Technology with hands-on experience developing UAV avionics, embedded systems, robotic mechanisms, custom electronics, and electromechanical prototypes. Experienced in hardware bring-up, system integration, PCB design, sensors, power electronics, microcontrollers, C/C++, Python, MATLAB, CAD, and iterative test and debug. Strong track record of building multidisciplinary systems from concept through bench and field testing.

## EDUCATION

**Rochester Institute of Technology — Rochester, NY**

- M.S. Electrical Engineering — Expected Dec 2026
- B.S. Electrical Engineering — 2026

## RESEARCH PROJECTS

**Autonomous UAV Perception & Avionics Platform — Research Assistant | Current**

- Develop and integrate an autonomous UAV platform combining flight-control hardware, Raspberry Pi 5, Hailo-8 AI acceleration, dual cameras, GPS, IMUs, telemetry, BLDC propulsion, and power electronics.
- Integrate and troubleshoot UART, I2C, CSI/MIPI, PCIe, networking, camera, sensor, and power interfaces during embedded hardware bring-up.
- Train and deploy custom YOLO vehicle-detection models for real-time onboard inference, achieving approximately 30 FPS on the embedded platform.
- Develop dual-camera detection and tracking software with persistent object IDs, distance/velocity estimation, visualization, and real-time video streaming.
- Integrate GPS, IMU, barometric, camera, and ArduPilot data for aircraft state estimation and target geolocation; analyze flight-test results to improve repeatability and localization accuracy.
- Develop Python/Linux tools for hardware diagnostics, sensor integration, data collection, performance monitoring, and system validation.

**Modular Robotic System for Collaborative Structures — M.S. Thesis | Current**

- Design and develop modular robotic systems capable of articulation, docking, moving, and lifting other modules to construct larger structures.
- Design gearboxes, drivetrains, shafts, bearings, structural components, and docking mechanisms using iterative CAD and rapid prototyping.
- Develop and evaluate bevel, cycloidal, and hypocycloidal gearbox concepts to improve torque transmission and lifting capability.
- Integrate ESP32 microcontrollers, stepper motors, motor drivers, sensors, and power electronics into electromechanical prototypes.
- Develop embedded C/C++ and ROS2/micro-ROS control architecture for distributed robotic operation.
- Diagnose mechanical binding, backlash, alignment, motor torque, wiring, power, and communication failures through iterative bench testing and redesign.

**Integrated Robot Controller PCB — Independent Engineering Project | Current**

- Design a custom robot-controller PCB integrating an ESP32, dual DRV8825 stepper-motor drivers, servo interfaces, USB-C programming, external GPIO, and onboard power conversion.
- Develop the power architecture with buck regulation, reverse-polarity protection, replaceable fuse protection, TVS/ESD protection, bulk capacitance, and filtering.
- Perform schematic capture, component selection, interface design, and PCB development in Altium Designer.
- Design interfaces for NEMA 17 stepper motors, servos, sensors, and external power while considering current capacity, protection, testability, and manufacturability.

## SKILLS

**Programming & Embedded:** Python, C, C++, MATLAB, VHDL, Verilog, ROS1, ROS2, micro-ROS, Embedded Linux
**Embedded Hardware:** ESP32, STM32, Raspberry Pi 5, Hailo-8, Microcontrollers, Sensors, Motor Drivers, Flight Controllers
**Electrical & Avionics:** Altium Designer, PCB Design, Schematic Capture, Power Electronics, Power Distribution, System Integration
**Interfaces:** UART, SPI, I2C, USB, Ethernet, MIPI/CSI, PCIe
**Test & Debug:** Oscilloscopes, Digital Multimeters, Bench Power Supplies, Hardware Bring-Up, Electrical Troubleshooting, Data Collection, System Integration
**Mechanical & Prototyping:** SolidWorks, Fusion, Inventor, CAD, 3D Printing, Mechanical Assembly
**AI / Computing:** YOLO, Edge AI, Computer Vision, Real-Time Object Detection and Tracking

## PROFESSIONAL EXPERIENCE

**AI Database Development**
MetroTek Electrical Services — Easton, PA | June 2025 – Aug 2025
• Improved database interfaces and automated information-intake workflows.
• Led employee training on asset data acquisition.
• Gathered user feedback to refine application features.

**Electrical Engineering Co-op / CAD Operator**
M/E Engineering — Rochester, NY | Jan 2024 – Aug 2024
• Designed electrical systems for K-12 renovations, new builds, and energy projects.
• Performed lighting calculations for classrooms, libraries, and gyms.
• Coordinated with plumbing, HVAC, architectural, and contractor teams.

**Superconducting Process Integration Intern**
Northrop Grumman — Linthicum, MD | June 2023 – Aug 2023
• Processed wafers in a FAB environment.
• Used JMP statistical software to analyze process trends.
• Collaborated across teams to develop a script for efficient lot-history data manipulation and visualization.

**Supply Chain Work Study**
Lutron Electronics — Bethlehem, PA | June 2022 – Aug 2022
• Validated supply data across departments.
• Organized third-party software data for integration.
• Built an Excel Power Query interface to identify and sort inventory input errors.

**Digital Systems I Teaching Assistant**
RIT — Rochester, NY | 2023 – Present
• Ran lab sessions, held office hours, graded labs and reports, and supported students in digital systems concepts.

**E-commerce Entrepreneur**
MattsSalesFloorStuff — Bethlehem, PA | 2017 – 2025
• Operate an online retail business including sourcing, listing, logistics, and customer service.

## LEADERSHIP

**RIT Cross Country / Track — Varsity Athlete and Team Captain**